

18.06 Problem Set 9

This short pset is due **Wednesday**, November 9, since Friday is a holiday.

Problem 1 (5+5+5 points)

Suppose that A is a 3×3 matrix with eigenvalues $\lambda_1 = 1, \lambda_2 = -1, \lambda_3 = 2$ and corresponding eigenvectors x_1, x_2, x_3 .

- (a) Give eigenvectors and eigenvalues of $(A^2 - 3A + 4I)^{-1}$
- (b) For what value(s), if any, of the scalar μ is $B = A^2 - 3A + \mu I$ **singular**, which corresponds to B having one or more eigenvalues equal to _____.
- (c) $A^n x$ for large n is very nearly parallel to _____ unless x is _____.

Problem 2 (10 points)

Run `using LinearAlgebra` followed by running `eigvals(randn(3,3))` a few times.

Explain why a 3×3 real matrix A must have at least one real eigenvalue, and in fact this must be true for any $m \times m$ real matrix if m is _____ (most general answer).

Problem 3 (3+3+4+3+3 points)

Consider the following recurrence:

$$f_n = \frac{f_{n-2} - f_{n-1}}{2}$$

Suppose that we start it with $f_0 = 0$ and $f_1 = 1$. Then the first few terms in the sequence are:

$$f_0, f_1, f_2, \dots = 0, 1, -\frac{1}{2}, \frac{3}{4}, -\frac{5}{8}, \frac{11}{16}, -\frac{21}{32}, \frac{43}{64}, -\frac{85}{128}, \frac{171}{256}, -\frac{341}{512}, \frac{683}{1024}, -\frac{1365}{2048},$$

(a) Similar to the Fibonacci sequence from class, we can write this recurrence in matrix form:

$$\begin{pmatrix} f_n \\ f_{n-1} \end{pmatrix} = A \begin{pmatrix} f_{n-1} \\ f_{n-2} \end{pmatrix}$$

for what matrix A ?

(b) Therefore, write a formula for $f_n = \text{---}^T A^n \text{---}$: fill in the blanks with two column vectors.

(c) Find the eigenvalues of A and corresponding eigenvectors.

(d) From (c), what should the ratio f_n/f_{n-1} approach for large n ? Check that your prediction matches what the terms in the sequence above appear to be doing.

(e) Give an exact formula for f_n , using (b) and writing --- in the basis of --- .